

COMMUNITY BASED EPIDEMIOLOGICAL/STATISTICAL STUDIES OF MALE PATIENTS/DISEASES IN OUT-PATIENT- DEPARTMENT AT SHIFA-UL-MULK MEMORIAL HOSPITAL FOR EASTERN MEDICINE

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ABSTRACT

Brief Description: Shifa-Ul-Mulk Memorial Hospital for Eastern Medicine is the first and only Hospital in Pakistan, providing health care facilities through Unani/herbal System of Medicine. It is a 50-bedded Hospital providing free health care. It has well equipped Wards, Laboratory, Operation Theater, Labour room, etc. The Hospital was founded in 2000 and since, it is the only Health Care Center of >28 villages near the city of education, Bait-Ul-Hikmah.

Purpose: Statistical data based on number of patients, diseases, age groups, etc has long been needed in order to promote and elevate the health care facilities to the patients of the rural areas around the Bait-Ul-Hikmah. The objective of our study is to determine the prevalence of male diseases specially those, not responded to therapies of modern medicine. Also, we have an eye on the diseases which are occurring more frequently, so that we are able to target the etiology and better management of these diseases.

Results: 35, 216 male patients were seen at the OPD. 6,919 in the Year 2000, 9,144 in 2001, 9,693 in 2002 & 9460 in 2003. Number of patient's increases from 2000 to 2002 & decreases in 2003. Disease of GIT (34.47%) has the highest prevalence while Respiratory disease (22.89%) being the second. Age group of 18-45 years has got the largest numbers and most of the patients were married. Respiratory, skin & infectious diseases are more common in children.

Keywords: Male Outpatients- Disease frequency-Age distribution-Key facts.

YEAR 2000

Introduction

The study reports from January 2000 to December 2003. In the Year 2000, there is a total of 6919 male outpatients were seen at the OPD of Shifa-Ul-Mulk Memorial Hospital, at a rate of 577 patients/Month (Fig. A1). It has been obvious from the graph A1 that number of patients increases in the middle of the year (April-September), while there frequency decreases at the extremes of the calendar year (October-March). June is the Month that contains the highest number of patients i.e. 696, whereas the least number of patients were seen in December i.e. 436.

Diseases Frequency

Illnesses regarding GIT (35.04%) and respiratory system (23.62%) have got the highest frequency. The other diseases accounts for only 41.34%. From these ENT (2.16%) and endocrinal (1.74%) diseases have very minor contributions.

Infectious diseases, dermatological problems, urolethiasis and sexual weaknesses are also occurring quite frequently.

Age Distribution

Table-A1 shows some very interesting features. For the sake of convenience, we divide the patients into three groups according

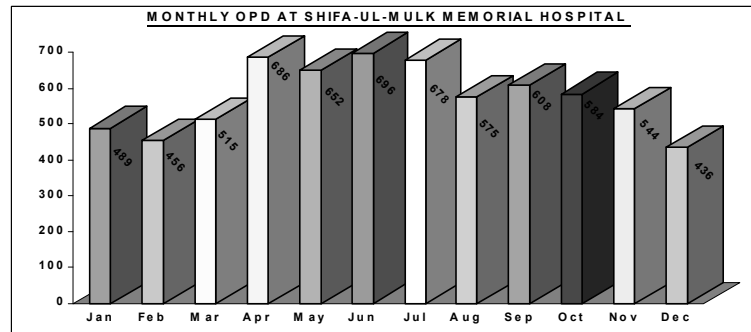


Fig. A1

Table-A1

Disease Pattern with Respect to Age Groups							
Diseases	Age Groups						Total %
	1-17 Years (Teenagers)		18-49 Years (Adults)		> 50 Years (Geriatric)		
	Patients	Percentage	Patients	Percentage	Patients	Percentage	
GIT	486	20.05%	995	41.05%	943	38.91%	35.04%
RTI	689	42.17%	444	27.18%	501	30.67%	23.62%
Urogenital	33	7.86%	261	62.15%	126	30%	6.07%
Skin	196	47.58%	167	40.54%	49	11.90%	5.96%
Fever	203	42.65%	160	33.62%	113	23.74%	6.88%
Weakness	34	10.12%	121	36.02%	181	53.87%	4.86%
CVS	02	0.84%	77	32.09%	161	67.09%	3.47%
CNS	31	9.39%	117	35.46%	181	55.16%	4.77%
Musculoskeletal	11	2.92%	147	43.77%	219	53.32%	5.45%
ENT	49	32.89%	39	26.18%	61	40.94%	2.16%
Endocrinology	03	2.47%	36	29.8%	82	67.77%	1.74%
Total	1737	25.10%	2564	34.64%	2618	37.57%	

to their ages, simply 1-17 years, 18-49 years, and more than 50 years.

Prevalence of respiratory, skin and infectious diseases are more common in teenagers (1-17 Years). In young and late adult patients, frequency of diseases regarding GIT and urogenital system are more common, while in the third group (geriatric), CVS, CNS, generalized weaknesses, musculoskeletal and endocrinal diseases are mainly occurring diseases.

Mostly the patients are married 52.27%, which makes a total of 3616 Marries Patients.

The rest of the patients are Unmarried and their total number is 3303.

Key Facts

After a follow through of the study of the Year 2000, it is quite clear that the Gastrointestinal Tract is the most common system to be affected and Respiration to be the second.

Third Place is occupied by Infectious Diseases. When we talk about the age groups with respect to diseases, Geriatrics (<50) (38%) has the highest prevalence, but they are very close to the Adults and post adults (18-

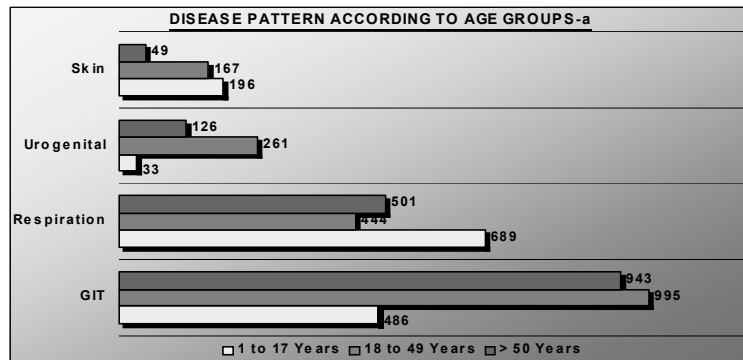


Fig.A2



Fig.A3

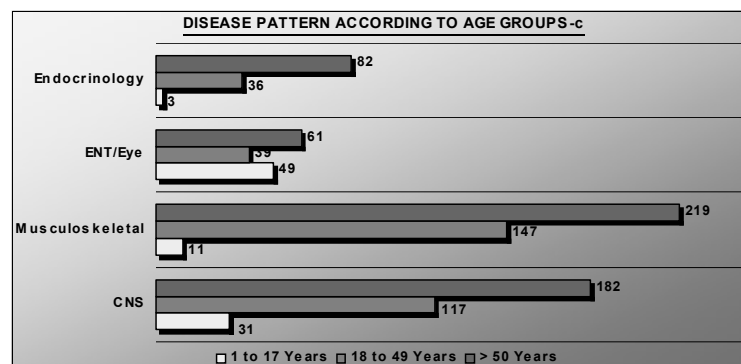


Fig.A4

49) (37.5%). Pediatric (1-17) lie on the third place (25.10%).

Mostly the patient came in the month of June (696) & highest number of patients came from April to July. The possible etiology of this will be discussed in the final conclusion.

YEAR 2001

Introduction

Year 2001, started in the same pattern, as 2000 i.e. mostly the patients are less at the beginning and at the end of the year. Total of 9144 patients were seen at the OPD of Shifa-

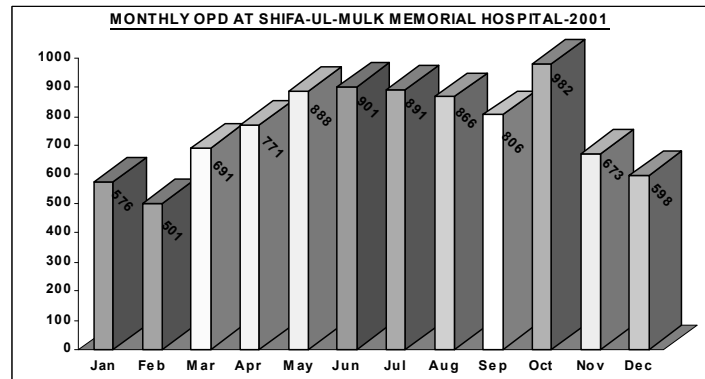


Fig.B1

Table-B1

Diseases Patterns with Respect To Age Groups							
Diseases	Age Groups						Total
	1-17 Years (Teenagers)		18-49 Years (Adults)		> 50 Years (Geriatrics)		
	Patients	%	Patients	%	Patient	%	
GIT	772	24.93%	1448	46.75%	877	28.31%	33.87%
RTI	1021	48.96%	453	21.73%	611	29.31%	22.81%
Urogenital	96	15.51%	341	55.08%	182	29.4%	6.77%
Skin	290	44.92%	221	33.53%	148	22.45%	7.21%
Fever	359	42.33%	303	35.74%	186	21.94%	9.28%
Weaknesses	55	13.81%	133	33.42%	210	52.76%	4.36%
CVS	04	1.49%	102	38.21%	161	60.30%	2.92%
CNS	35	8.21%	130	30.52%	261	61.27%	4.66%
M/S	11	2.55%	147	34.10%	273	63.35%	4.72%
ENT/Eye	53	31.54%	44	26.19%	70	41.67%	1.84%
Endocrinology	01	0.69%	57	39.04%	88	60.27%	1.60%
Total	2697	29.49%	3379	36.97%	3067	33.55%	

Infections/Fevers also occurring quite regularly (848 patients, 9.28%). The other diseases accounts for only 34.05%, among which, diseases related to Endocrinology and ENT are much less.

Ul-Mulk Memorial Hospital at a rate of 762 patients/Month. This figure is quite high as Month of November i.e. 982 patients (Fig. B1). June contains the second highest numbers and the least number of patients came in the Month of February.

Disease Frequency

The same pattern has been followed over here also. 3097 (33.87%) patient belongs to

Gastroenterology, whereas 2085 (22.18%) related to Respiratory System (Fig. B2). It means that more than 56% of patients came due to the illnesses regarding either GIT or Respiration.

Age Distribution

Unlikely from the year 2001, mostly the patients belongs to the age group of Adults and Post adults (18-49), they together makes 3379

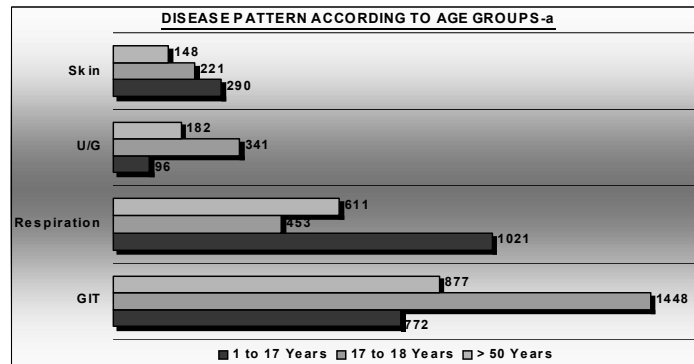


Fig.B2

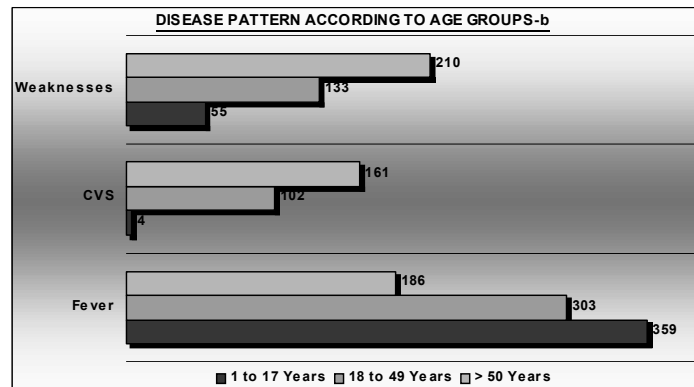


Fig.B3

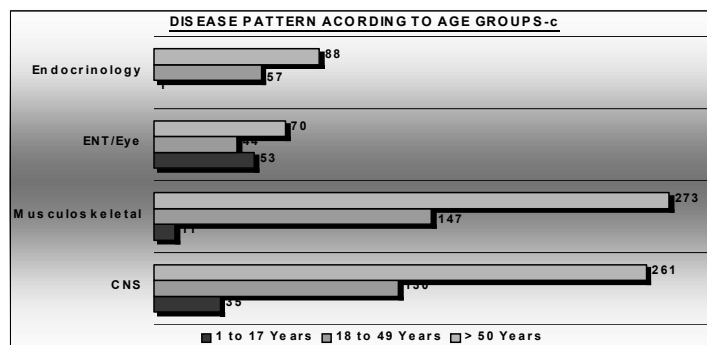


Fig.B4

Patients at an average of 36.97%. Frequency of pediatric diseases also increases from 25.10% to 29.49% and it makes a 2697 pediatric patients. Age Group of more than 50 accounts for 3067 Patients with an average of 33.55% (Table-B1).

Diseases related to Respiration, Skin and infections are more common in the pediatrics (Fig. B3).

Endocrinal and musculoskeletal diseases, diseases of CNS and CVS and weaknesses are more common in old age (Fig. B4).

Urogenital which also includes the sexual weakness is more common amongst the age group of 18-49 years. Married Patients have an average of 53.41% and their total number is 4884 (Fig. B2).

Key Facts

In the year 2001, Gastro-intestinal System is the most frequently affected system of the people of the villages around the Bait-UL-Hikmah. Respiratory System has second highest prevalence, while infectious diseases being the third.

from 25.10% to 29.49% this year and makes a total of 1697 Patients. Old Patients (>50) are decreased, now they are 33.5% from 38%. The remaining are the adults and post adults (36.97%), which has the highest prevalence. Highest Number of Patient came in October (982) and mostly the patient came to visit OPD from May to October. Married are patients are more as compare to unmarried. This completes our study of the Year 2001.

YEAR 2002

Introduction

Number of Pediatric Patients increases

Year 2002 contains the highest number of

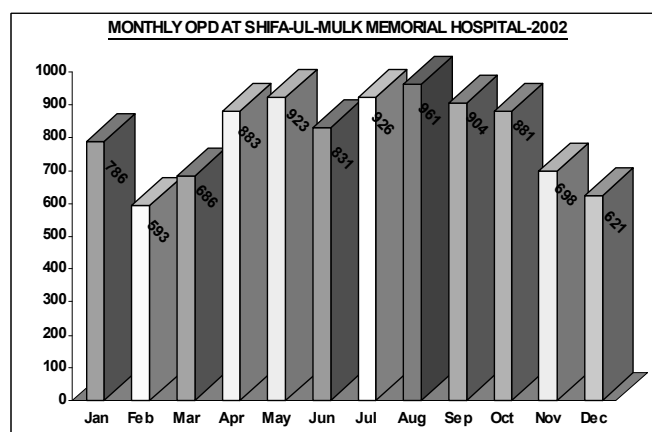


Fig.C1

Table-C1

Disease Pattern According to Age Groups							
Diseases	Age Groups						Total
	1-17 Years (Teenagers)		18-49 Years (Adults)		> 50 Years (Geriatric)		
	Patients	%	Patients	%	Patients	%	
GIT	761	23.42%	1603	49.33%	886	27.26%	33.53%
RTI	991	45.72%	511	23.57%	666	30.71%	22.36%
Urogenital	66	11.92%	327	59.02%	161	29.06%	5.72%
Skin	231	36.16%	255	39.91%	153	13.94%	6.60%
Fever	611	51.18%	383	32.08%	200	16.76%	12.32%
Weaknesses	46	10.85%	167	39.39%	211	49.76%	4.38%
CVS	03	1.05%	104	36.11%	181	62.84%	2.98%
CNS	40	9.11%	151	34.40%	248	56.50%	4.53%
M/S	08	1.47%	237	43.40%	301	55.12%	5.64%
ENT/Eye	39	28.05%	44	31.65%	56	40.28%	1.44%
Endocrinology	01	5.77%	18	34.61%	33	63.47%	0.55%
Total	2797	28.85%	3800	39.20%	3096	31.94%	

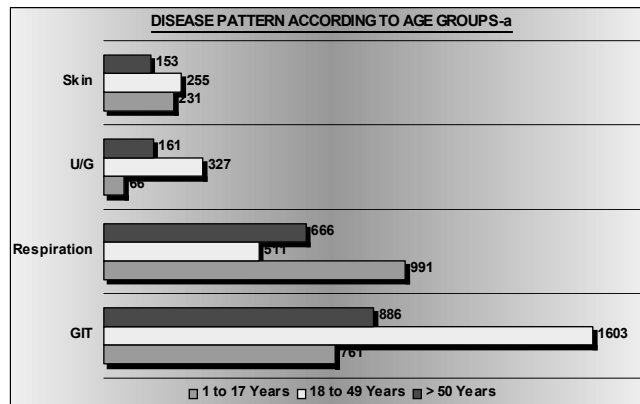


Fig.C2

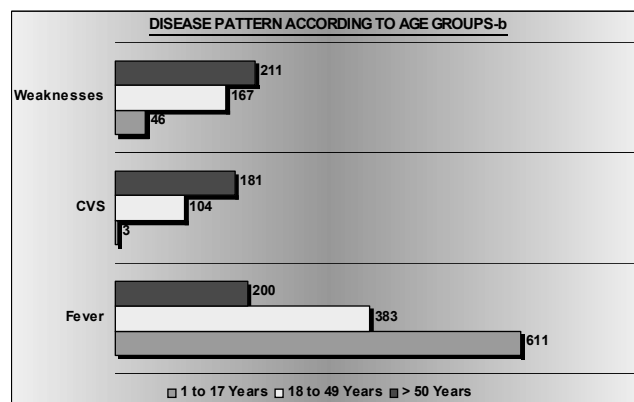


Fig.C3



Fig.C4

patients so far. The number of male patients reaches 9693 during this year at an average of 808 Patients/Month. From May to September, there is highest number of Patients (Fig. C1).

Highest number of Patients is in the month of August i.e. 961. In February, 593 patients visited OPD which is the least number of the year. The graph is quite evenly balanced

when compares from the last two years. Similar to the last Year, frequency of Patient is less at the beginning and at the end of the year.

Disease Frequency

Number of Gastro-intestinal Patients increases up to 3250, which again makes this system to be the highest frequency of patients viz. 33.53%. Respiratory System again follows GIT, 2168

A patient belongs to it. Infectious diseases increases quite tremendously this year, they reached almost 1200 at an average of 12.32%. Again diseases related to Endocrinology, ENT and CVS are much less this year.

Age Distribution

Most of the Patients lie between the age group of 18-49 Years (Adults and Post adults). Their total number is 3800 at an average of 39.20%, followed by the geriatrics (31.94%). Pediatric patients almost remain the same-28.88% (Table-C1). Respiratory, dermatological and Infectious diseases are the most common cause of the pediatric patients to visit the OPD this year (Fig. C2).

Generalized weaknesses, diseases of CVS, CNS, endocrinal and musculoskeletal diseases are more common in Old Patients (Fig. C4). Gastrointestinal and urogenital diseases are

more common in Adults and Post adults (Fig.C3). Numbers of Married Patients are much more during this year. 54.58% Patients are married. Number of Unmarried Patients is 4403.

Key Facts

Year 2002 has almost the same statistics as previous two years. Diseases regarding Gastro-intestinal and Respiratory Systems and Infectious Diseases exist with the percentage of 33.53%, 22.36% and 12.32% respectively, accounts for almost 70% of the diseases. Number of Pediatric Patients reaching 2797, Geriatrics include 31.94%. In the figure below most of the space is occupied by Adults and Post adults, which have the highest frequency as far as the number of patients are concern (39.20%). Mostly patients came in the mid of the year. The most important thing is that this year has the highest number of male patients and most of them are belong to the age group of 18-49 Years.

YEAR 2003

Introduction

In the Year 2003, there is a total of 9460 Patients visited the OPD at Shifa-Ul-Mulk Memorial Hospital, at an average of 789 Patients/Month. The number of Patients decreases as compare to the last year and this

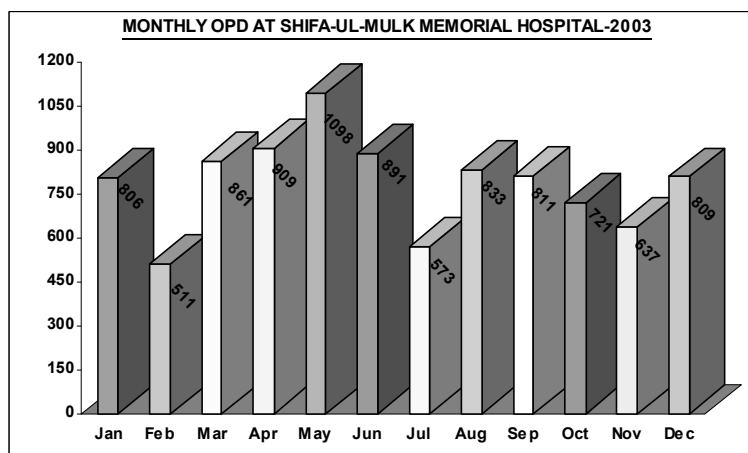
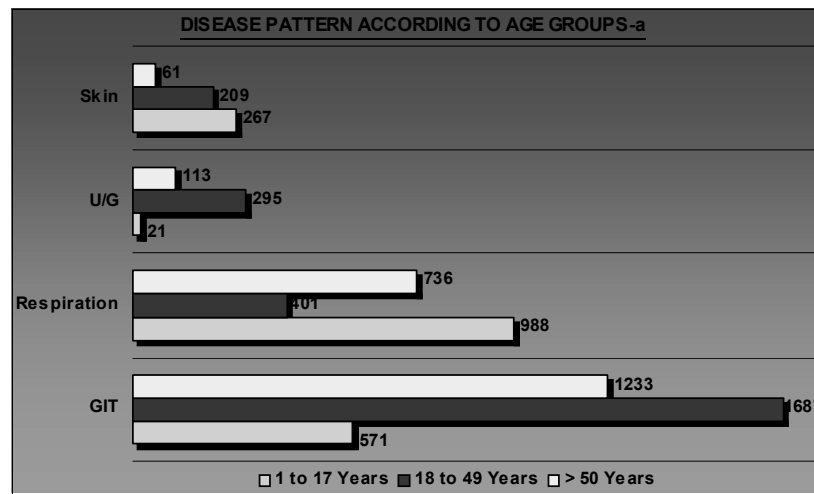


Fig.D1

Table-D1

Disease Pattern According to Age Groups							
Diseases	Age Groups						Total
	1-17 Years (Teenagers)		18-49 Years (Adults)		> 50 Years (Geriatric)		
	Patients	%	Patients	%	Patients	%	
GIT	571	16.35%	1687	48.32%	1233	35.31%	36.91%
RTI	988	46.5%	401	18.88%	736	34.63%	22.47%
Urogenital	21	4.9%	295	68.77%	113	26.34%	4.45%
Skin	267	49.71%	209	38.91%	61	11.35%	5.68%
Fever	491	44.8%	218	19.9%	387	35.31%	11.59%
Weaknesses	22	5.08%	110	25.40%	301	69.51%	4.58%
CVS	NIL	----	73	26.94%	198	73.06%	2.86%
CNS	19	6.01%	86	27.13%	212	66.88%	3.35%
M/S	08	1.35%	192	32.54%	390	66.10%	6.24%
ENT/Eye	53	44.16%	28	23.34%	39	32.5%	1.27%
Endocrinology	NIL	----	17	33.33%	34	66.67%	0.54%
Total	2440	25.8%	33160	35.05%	3704	39.15%	

**Fig.D2**

happens for the first time. Otherwise the number of Patients increasing gradually from the Year 2000 to the Year 2002. Mostly the Patients came in the first Semester of the year, Unlikely from the last three years. In May there is a total number of 1098 Patients (Fig.D1). This is the highest Number of Patient in any Month, in our whole study. In

February, there are 511 Patients and thus it has the shortest height in the graph.

Mostly the Patients came in the first Semester of the year, Unlikely from the last three years. In May there is a total number of 1098 Patients (Fig. D1). This is the highest Number of Patient in any Month, in our whole

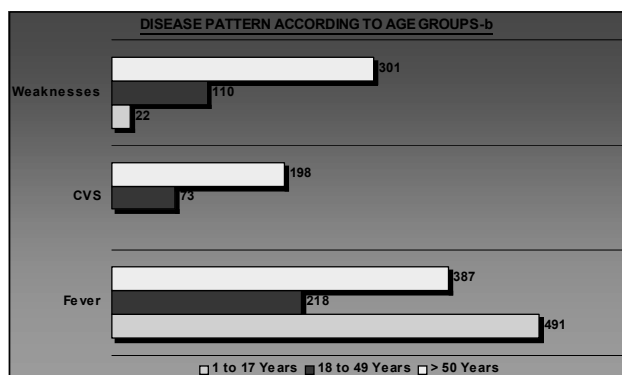


Fig.D3

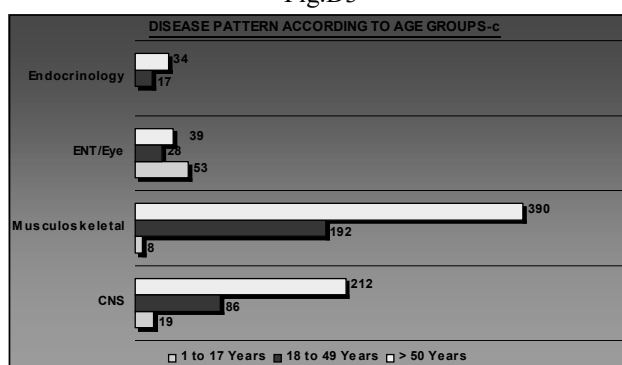


Fig.D4

study. In February, there are 511 Patients and thus it has the shortest height in the graph (Fig. D1).

Age Distribution

Unlike from the last Year, number of Old Patients (>50) or Geriatric is the highest i.e. 3704 at an average of 39.15% (Table D1). Pediatric Patients also decreases to 25.8%.

Adults and Post adults also decreases slightly. Still Respiratory, Infectious, and Skin diseases are more common amongst the age groups of 1-17 Years (Fig. D3).

In Geriatric, diseases of CVS, CNS, Endocrinal and Musculoskeletal diseases are more common (Fig. D4). In the age groups of 18-49, frequency of Urogenital and GIT diseases are more common (Fig. D5). Mostly the Patients are married. A total of 5731

patients are married (59.45%) and 40.55% patients are not married.

Key Facts

This Year (2003) is quite different as far as statistics are concerned. It does not follow the same pattern 100%. As was previous.

Although, GIT, Respiratory System and Infectious diseases have the highest frequency, GIT has the highest number of Patients than any year.

As it is quit clear number of Adult and Post adult as well as Pediatric patient decreased (35.5% and 25.8% respectively) and hence Old Patient increases during the year (39.91%). In August of this Year, highest number of patient came to the OPD at Shifa-UI-Mulk. Now let's turn to the comparative

analysis of the four years as well as the final conclusion of the study.

COMPARATIVE ANALYSIS OF OPD FROM 2000-2003

Introduction

From 2000-2003, there is a total of 35,216 Male Patients. In the year 2000 there are 6919 patients and this figure increases to 9144 in 2001. In 2002 the figure again increases to 9693 and in 2003 the number of patients decreases to 9460. This shows that, patient's increases from 2000 to 2002. There is more than 20% increase in the OPD of Shifa-UI-Mulk from 2000 to 2001. Year 2002 has got the highest figure i.e. 9693, while 2000 has the lowest (6919).

The reasons, that why the OPD was dynamically increased from 2000 to 2001, is that there was two days off and the Hospital works in the working hours of the Hamdard University. But when the facility was extended to 24 hours a day increases during the year (39.91%). In August of this Year, highest number of patient came to the OPD at Shifa-UI-Mulk.

Age Distribution

When we categorize the patients into three age groups (1-17 Years, 18-49 Years and > 50 Years), we came to know that in 2000, there are 1737 (25.10%) patients belongs to the first age group i.e. *Pediatrics*. 2564 (37.09%) Male patients were either *Adults* or *Post adults*;

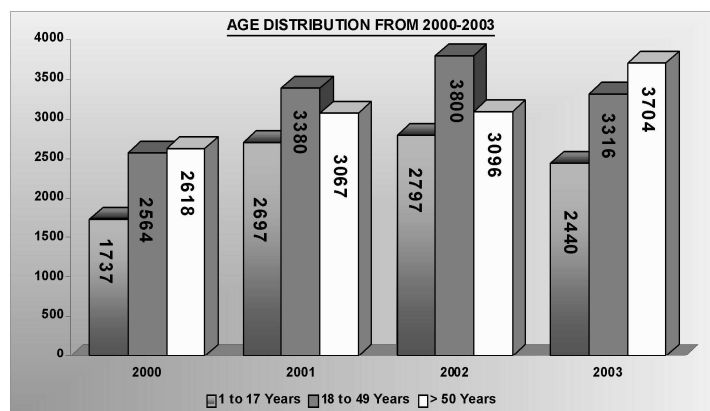


Fig.E1

Table-E1

Annual Distribution of Diseases						
Diseases	2000	2001	2002	2003	Total	%
GIT	2424	3097	3250	3491	12262	34.81%
Respiration	1634	2085	2168	2125	8012	22.75%
Infections	476	848	1194	1096	3614	10.27%
Skin	412	659	639	537	2247	6.39%
Urogenital	420	619	554	429	2022	5.74%
Musculoskeletal	377	431	546	590	1944	5.52%
Weaknesses	336	398	424	433	1591	4.51%
CNS	330	426	439	317	1512	4.30%
CVS	240	267	288	271	1066	3.02%
ENT/Eye	149	168	139	120	576	1.71%
Endocrinology	121	146	52	51	370	1.05%

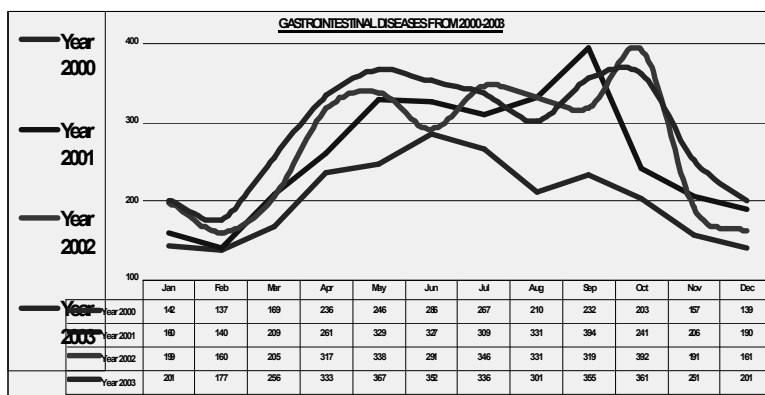


Fig.E2

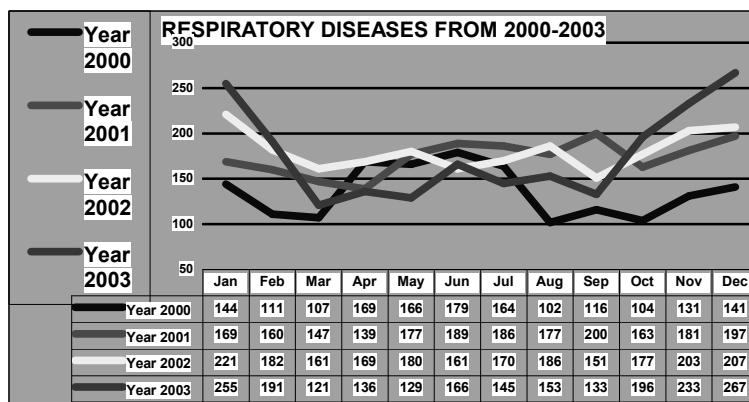


Fig.E3

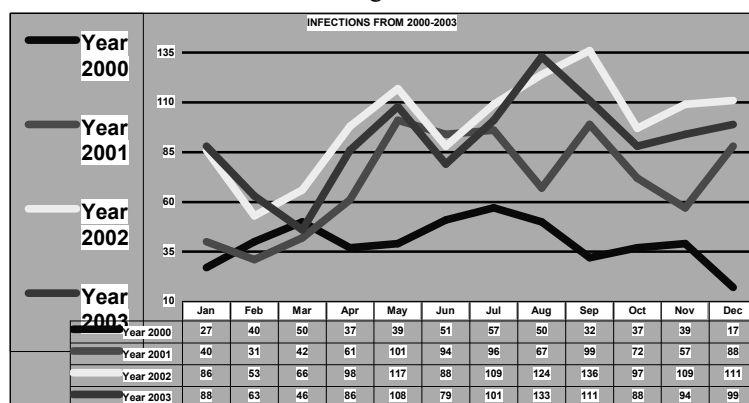


Fig.E4

whereas 37.8% (2618) patients are of age more than 50 (Fig. E1). In the Year 2000 and 2003, most of the patients belong to the age group of > 50 and in 2001 and 2002 adults and post adults have the highest prevalence of

diseases. However, the pediatric group always has the least quantity of patients.

The highest number of patients belong to any age group is 3800 in the year 2002, and

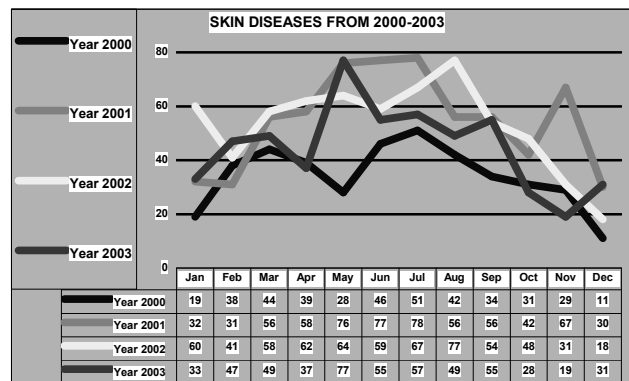


Fig.E5

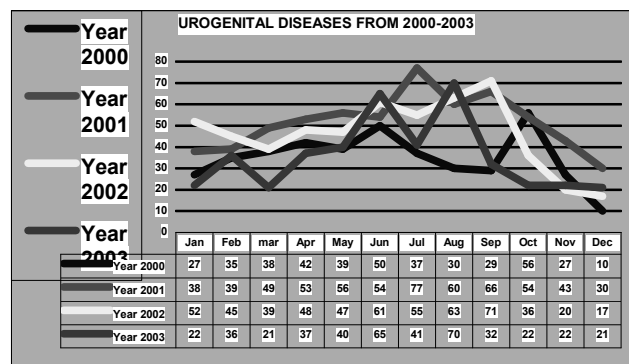


Fig.E6

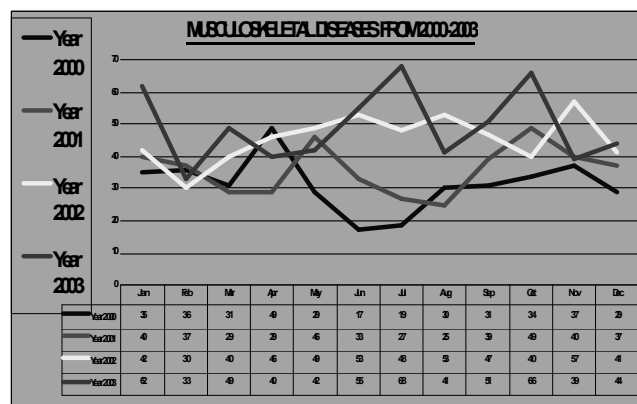


Fig.E7

that age group is of adults and post adults. Likewise, the lowest figure is of 1737 i.e. of Pediatric patients.

This shows that most of diseases were encountered in age groups of more than 17

years. Although diseases of pediatrics are less common, it does not mean that these diseases are occurring less frequently. The reason that why patients of age group > 50 are less than adults and post adults is that those patients who need surgery are not included.

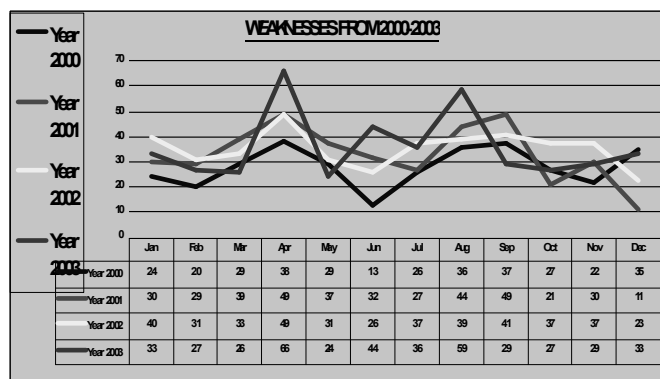


Fig.E8

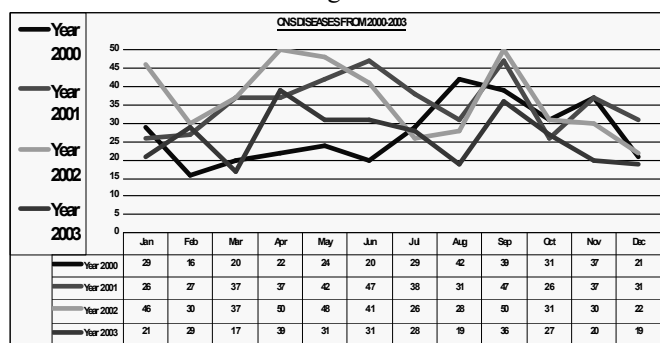


Fig.E9

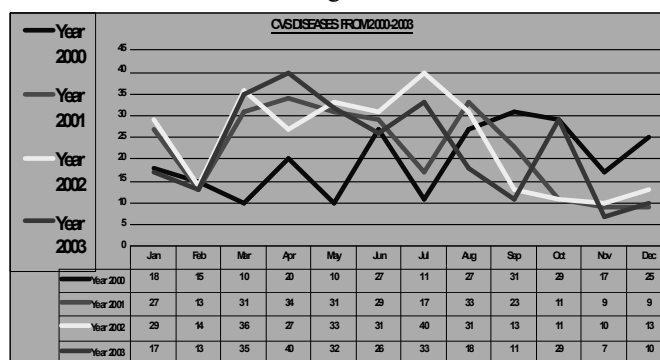


Fig.E10

Disease Distribution

Table E1 shows the total number of patients per year according to the disease pattern, which has been arranged in descending order. According to this table, diseases of *GIT*, *Respiration* and *Infectious diseases* are the commonest cause of the patients to visit the OPD. *CVS*, *ENT/Eye* and *endocrinal diseases* are less common. Here

again it proves that diseases, which require surgeries, are less common.

From here on, we'll discuss the diseases pattern individually.

Gastrointestinal Diseases

Pattern of diseases regarding *GIT*, which also includes the Liver and Billiary Tract have

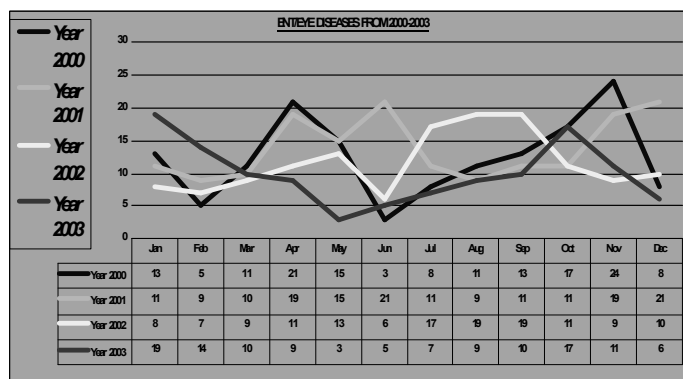


Fig.E11

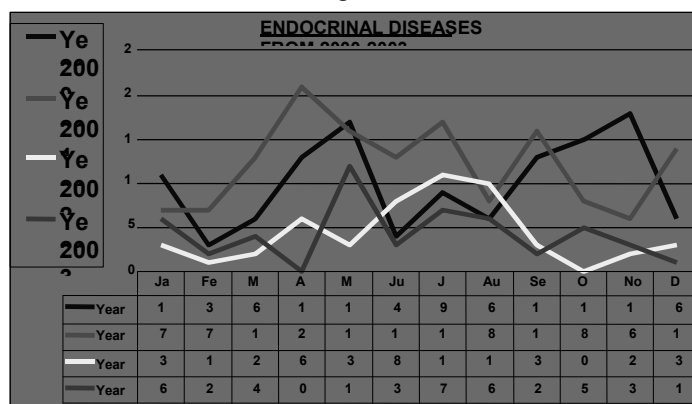


Fig.E12

been shown in the Figure-E1. Mostly the diseases are Stomatitis, Gastritis, Peptic Ulcers, Constipation and Hemorrhoids. Also Hepatitis, CLD, Pancreatitis and Jaundice also occurring quite frequently. In Pediatric Patients, Intestinal Worms is one of the major contributions.

This system has the highest prevalence amongst all other diseases (Table-E1). Graph (Fig. E2) shows that Patients visiting the OPD mainly in the Middle of the Year due to the ailments of GIT.

Respiratory Diseases

In Pakistan, winter season starts from November-December and remains till January-February. During this season the diseases like Upper Respiratory Tract Infections,

pneumonia (in Children), chronic obstructive pulmonary diseases (COPD) like Asthma and Bronchitis, occurs more frequently. This statement is proved by the graph below (Fig.E3). Patients are more in number from November to February. Mostly the age group involved is 1 to 17 Years. Respiratory System has the second highest prevalence and Pulmonary Tuberculosis has some serious contribution to it.

Infectious Diseases

It includes different infections in which there is high-grade fever. Mostly the infections are viral and they are very common among children. These diseases are less common in first trimester of the Year i.e. from January to March (Fig. E4). Measles is very common amongst the children.

Skin Diseases

Skin Diseases are also very common in Pediatric out Patients department. This includes Dermatitis, Eczema, Vitiligo, Psoriasis acne, Seborrhoea, and Skin Allergies etc. They occur quite frequently and regularly throughout the year (Fig. E5).

Urogenital Diseases

Urogenital Diseases includes Upper and Lower Urinary Tract Infections like Acute and Chronic Pyelonephritis, Urethritis, Cystitis, Urethritis, and Prostatitis.

Hydronephrosis, Urolethiasis, and other Urinary Complaints also come under this heading. But, the major contribution is of male Genital Diseases, Like Impotency, Decreased Libido, Erectile Dysfunction and Nocturnal Emission. The disease has highest frequency among the Adults and Post adults. The graph (Fig. E6) is quite evenly balanced which means that patients of this group used to come throughout the year regularly.

Musculoskeletal Diseases

Diseases like Arthritis, Lumbago, Schiatica, Gout, Spondylitis, and Osteomyelitis etc lie in the group of Musculoskeletal Diseases. It follows the fact that these are more common in patients of age > 50 Years. (Fig. E7)

Generalized Weaknesses

Although it is not a disease can give rise to many diseases. So it is very important to treat it as a major Problem. We find many patients that came under this heading every year (Fig. E8). Weaknesses are very common in old aged patients due to malnutrition and overload, which is further aggravated by unhygienic condition present all around their environment.

Central Nervous System

CNS diseases are very common in old aged patients and these include Paralysis, Migraine, Headache, Epilepsy, Meningitis, and TIA. We also include Psychiatric Diseases like Depression, Anxiety, Schizophrenia,

Dementia, Mania and Personality Disorders. The Graph (Fig. E9) is smoothly regular. Migraine, Paralysis and Depression are very common occurring diseases of CNS.

Cerebrovascular System

Ischemic heart diseases as Angina and Myocardial Infarction and Atherosclerotic Heart Diseases are very common diseases globally. But here in our study, its contribution is much less, as most of the coronary artery diseases required surgery (CABG). Awareness regarding Heart diseases also seems to be a contributing factor in less number of CVS patients. The Graph (Fig. E10) shows how they occur at Shifa-UI-Mulk Memorial Hospital from 2000-2003.

Ear, Nose and Throat/Eye Diseases

It also includes a large number of diseases like Tonsillitis, pharyngitis, Otitis Media, DNS, Rhinitis, Conjunctivitis, Cataract, Blephritis, Ectropion, etc. Their numbers are very less and these diseases occupy the second last position in our selected groups of diseases. Diseases like Chronic Tonsillitis and Eye diseases that needs surgery (Cataract) does not included in this section. These diseases follow no pattern in the graph (Fig. E11).

Endocrinal Diseases

Diabetes Mellitus is one of the alarming diseases in our country and is very common all over the world. But in the villages near Hamdard University, people have almost no awareness about these diseases. That's why there number is less at the OPD of Shifa-UI-Mulk and this group comes in the last of all. The other diseases that includes in this group are diseases of thyroid (Hypothyroidism, Myxedema, thyroiditis), pituitary and other endocrinal diseases. Below is the annual graph (Fig. E12).

Marital Status of the Patients

Figure-E14 shows that 55% of the patients visiting the out patient department (OPD) at Shifa-UI-Mulk Memorial Hospital of Eastern Medicine are married. It also includes widowed patients. But, the high frequency of

married patients is mainly because of the young age marriages that are practicing in the villages. The frequency of diseases in female patients is more common due to this fact, but it is not a part of our study.

CONCLUSION

The study comprises a total of 35216 male patients (in our selected groups of diseases) visited the OPD at Shifa-Ul-Mulk Memorial Hospital from the Year 2000 to 2003. In 2002, there is highest number of patients i.e. 9693. In June 2003 there is highest number of patients i.e. 9693.

Most of the patients belong to GIT, Liver and Billiary Tract diseases (12262 enrolments). Respiration and Infectious diseases are also very common. Diseases of CVS, ENT and endocrinology are occurring less commonly. Age Groups of Adults and Post adults have the highest frequency of diseases (13060 enrollments), but geriatrics lie very close to them. Pediatric Patients are less common, but they have major contribution in

Respiratory, skin and infectious diseases. Mostly the patients are married, 55% of the patients, visiting OPD are married, and their number reaches 19521 enrollments (Out of 35216).

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